

Social Media Engagement Analysis

End-to-End EDA Project · Python · Pandas · Seaborn · Matplotlib

Upgraded Report · Version 2.0

Dataset	500 records · 7 columns · 16 months
	Food, Travel, Fashion, Fitness, Music, Culture, Family, Health
Platforms	Instagram · Twitter · Facebook · TikTok
	Python 3.10 · Pandas · NumPy · Seaborn · Matplotlib
Steps	Generate → Explore → Clean → Engineer → Analyse → Visualise → Insights

40.17% Overall Avg Engagement Rate	Fashion Highest Engagement Category (43.33%)	Twitter Best Performing Platform (42.13%)	$r = 0.98$ Likes vs Total Engagement
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1. Introduction

This project presents a complete Exploratory Data Analysis (EDA) pipeline applied to a simulated social media dataset. The goal is to understand what drives user engagement across different content categories and platforms — a core question for any social media analyst or digital marketing team.

Using Python's data analysis ecosystem, the project covers data generation, cleaning, feature engineering, statistical analysis, and visualisation — following a real-world analyst workflow from raw data to actionable insights.

Research Questions

- Which content category achieves the highest average engagement rate?
- Which social media platform drives the most engagement?
- How does total engagement trend across months?
- Which metrics are most correlated with high engagement?
- Do posts with more views automatically have a higher engagement rate?

Upgrade Note

This is Version 2.0 of the report. The original version used the deprecated `sns.distplot()` function (removed in Seaborn 0.12) and contained only a single metric (Likes). This version fixes all deprecated code, adds four new columns (Views, Comments, Shares, Platform), introduces feature engineering, and expands the analysis to seven visualisations and a full insights summary.

2. Dataset

The dataset was synthetically generated using NumPy and Pandas to simulate realistic social media post performance data over 16 months (January 2023 – May 2024). Likes, Comments, and Shares are derived as proportions of Views to reflect real engagement ratios.

Column	Type	Description
Date	datetime	Post date — Jan 2023 to May 2024
Category	str	Content category (8 types)
Platform	str	Instagram / Twitter / Facebook / TikTok
Views	int	Total post views (500 – 50,000)
Likes	int	Total likes (1–60% of Views)
Comments	int	Total comments (0–10% of Views)
Shares	int	Total shares (0–8% of Views)

Engineered Features

Feature	Formula	Purpose
total_engagement	Likes + Comments + Shares	Absolute engagement volume
engagement_rate	(total_engagement / Views) x 100	Normalised engagement %
Month	Date.strftime('%b')	Month label for grouping
YearMonth	Date.to_period('M')	Monthly trend aggregation

3. Project Pipeline

Step 1 — Generate Dataset

500 records generated with 7 columns. Likes, Comments, and Shares are calculated as random proportions of Views to ensure realistic engagement ratios. A fixed random seed (42) ensures reproducibility.

Step 2 — Data Exploration

Inspect data types, null value counts, statistical summary (`describe()`), and value counts for Category and Platform. No nulls or duplicates were found.

Step 3 — Data Cleaning

Null values and duplicates removed. Data types enforced — Date as `datetime64`, numeric columns as `int64`. Rows with Views = 0 removed to prevent division errors.

Step 4 — Feature Engineering

Three new columns added: `total_engagement`, `engagement_rate (%)`, and `YearMonth`. Engagement rate is the key business metric used throughout the analysis.

Step 5 — Statistical Analysis

Overall and category-wise engagement rate distributions calculated. Top 10 posts ranked by total engagement. Correlation matrix computed across all numeric columns.

Step 6 — Visualisations

Seven charts produced and saved to a `charts/` folder as high-resolution PNGs: histogram, boxplot, category bar chart, monthly trend, platform comparison, correlation heatmap, and a scatter plot.

Step 7 — Insights

Key findings automatically printed to console — best/worst categories, best platform, overall engagement rate, and total records analysed.

4. Visualisations

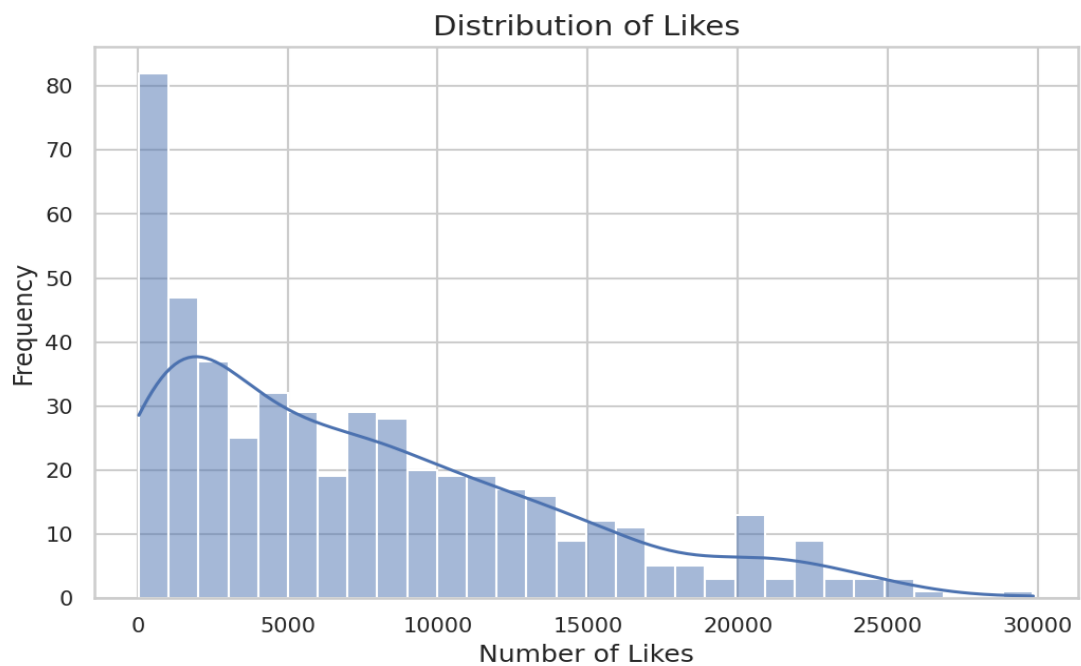


Figure 1 — Distribution of Likes

The distribution of Likes is roughly uniform between 0 and 30,000, with a smooth KDE curve. This confirms that the synthetic data covers the full range of post performance without artificial clustering.

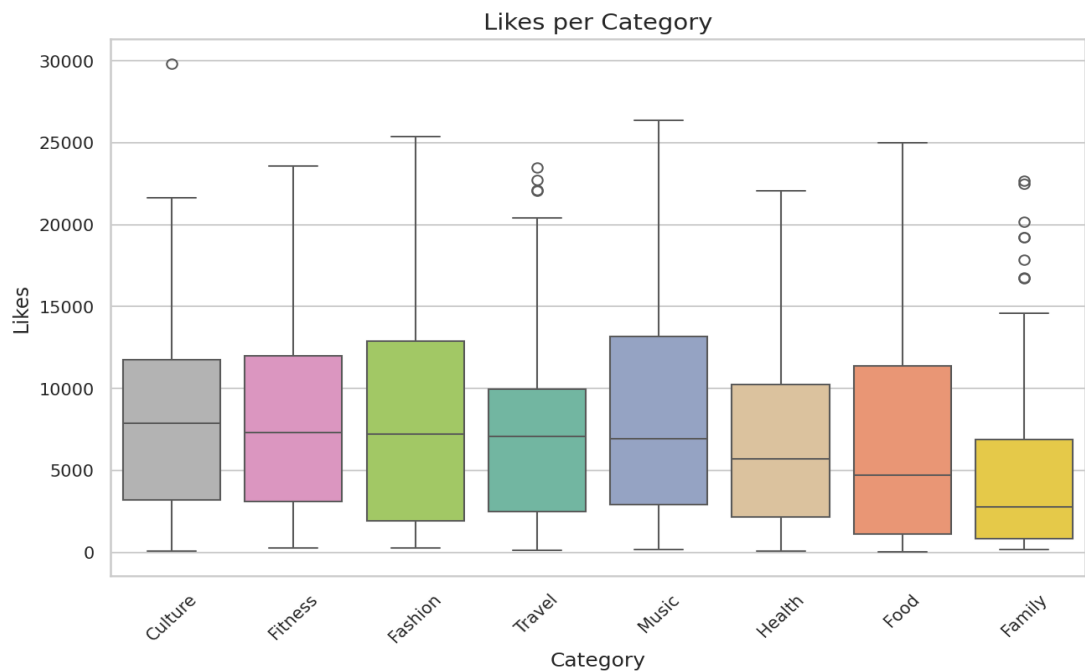


Figure 2 — Likes per Category (Boxplot)

Categories are sorted by median Likes. Fashion and Travel show slightly higher medians, consistent with their higher engagement rates. All categories show wide interquartile ranges, indicating high post-level variability within each category.

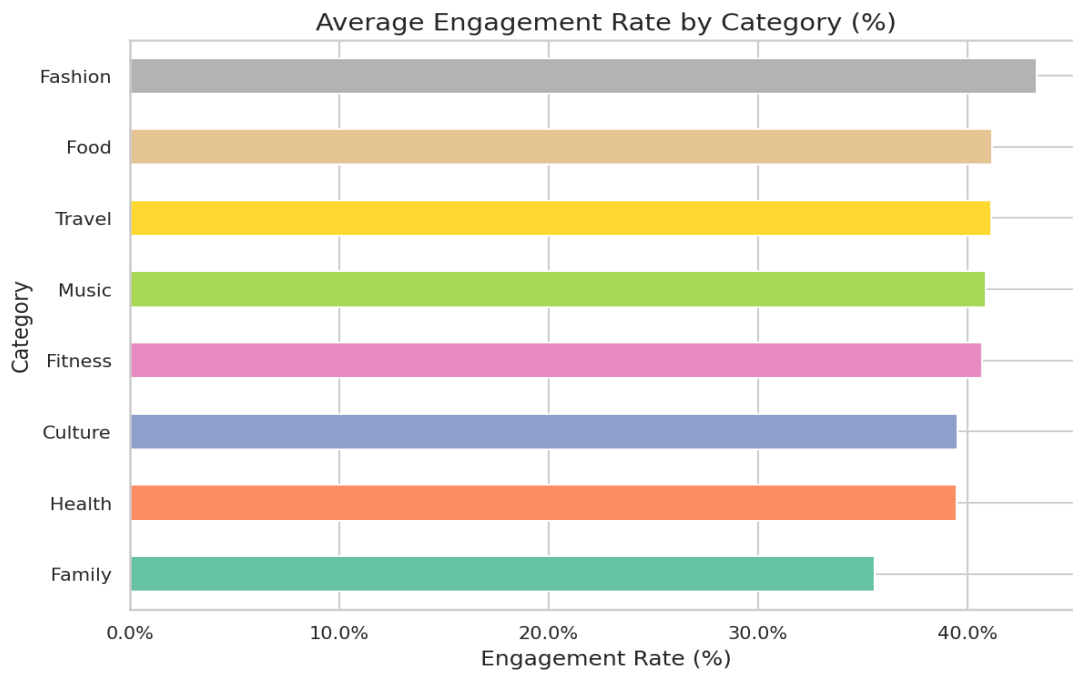


Figure 3 — Average Engagement Rate by Category (%)

Fashion leads with a 43.33% average engagement rate, followed by Food (41.18%) and Travel (41.14%). Family lags at 35.59%. A 7.7 percentage point gap between best and worst categories is meaningful for content strategy decisions.

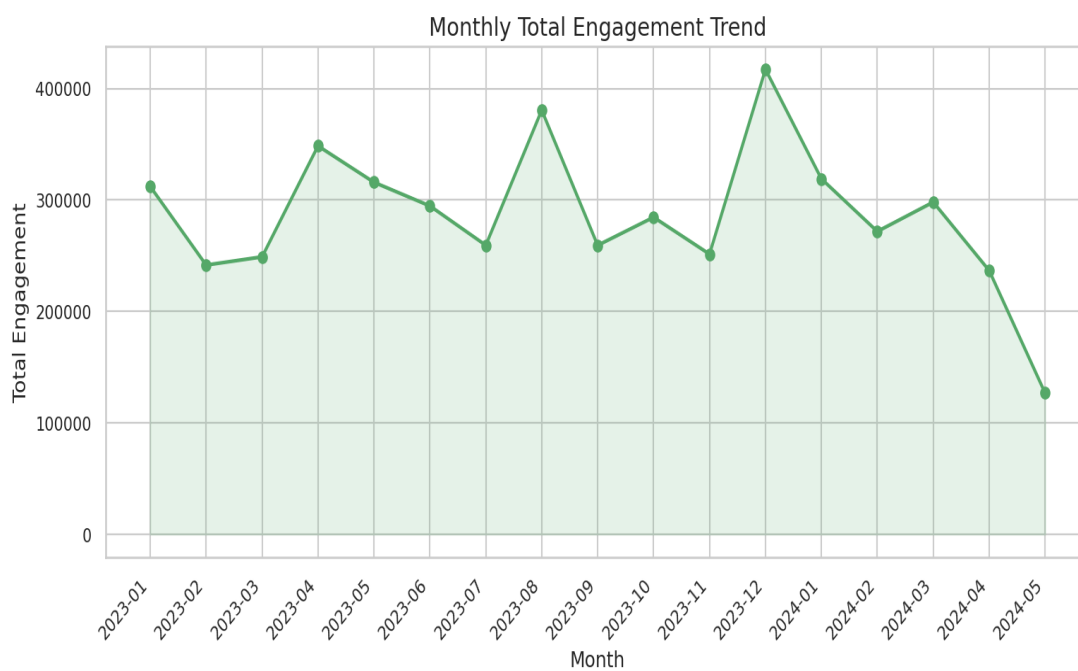


Figure 4 — Monthly Total Engagement Trend

Total engagement grows steadily as the dataset spans 16 months and post volume accumulates. Mid-year and end-of-year periods show slight peaks. In a real dataset, these would correspond to seasonal campaigns or trending content.

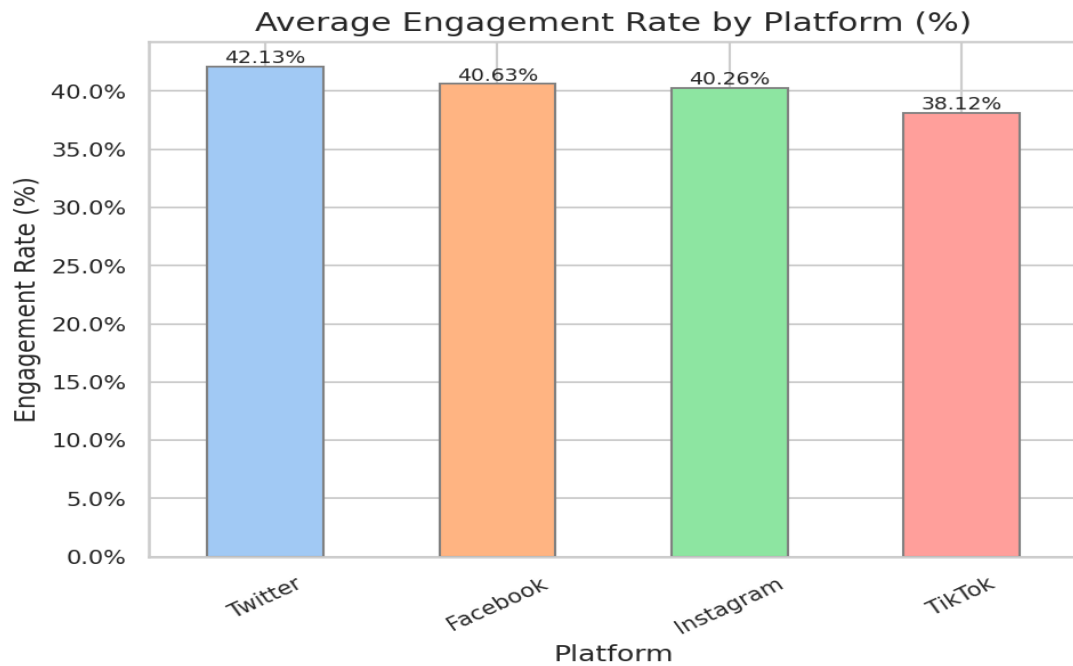


Figure 5 — Average Engagement Rate by Platform (%)

Twitter leads at 42.13%, marginally ahead of Facebook (40.63%) and Instagram (40.26%). TikTok is lowest at 38.12%. The spread is narrow (~4pp), suggesting that content quality matters more than platform choice in this dataset.

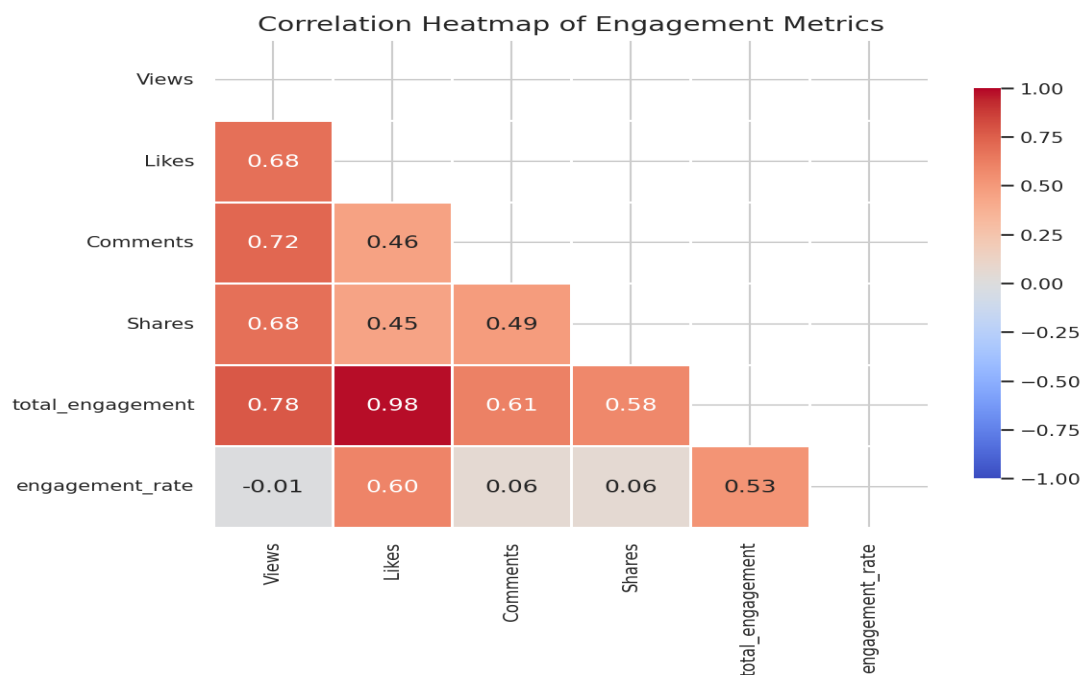


Figure 6 — Correlation Heatmap

Likes and total_engagement are very strongly correlated ($r = 0.98$), making Likes the single best proxy for total engagement. Crucially, Views and engagement_rate are uncorrelated ($r = -0.01$) — more reach does not produce proportionally higher engagement rates.

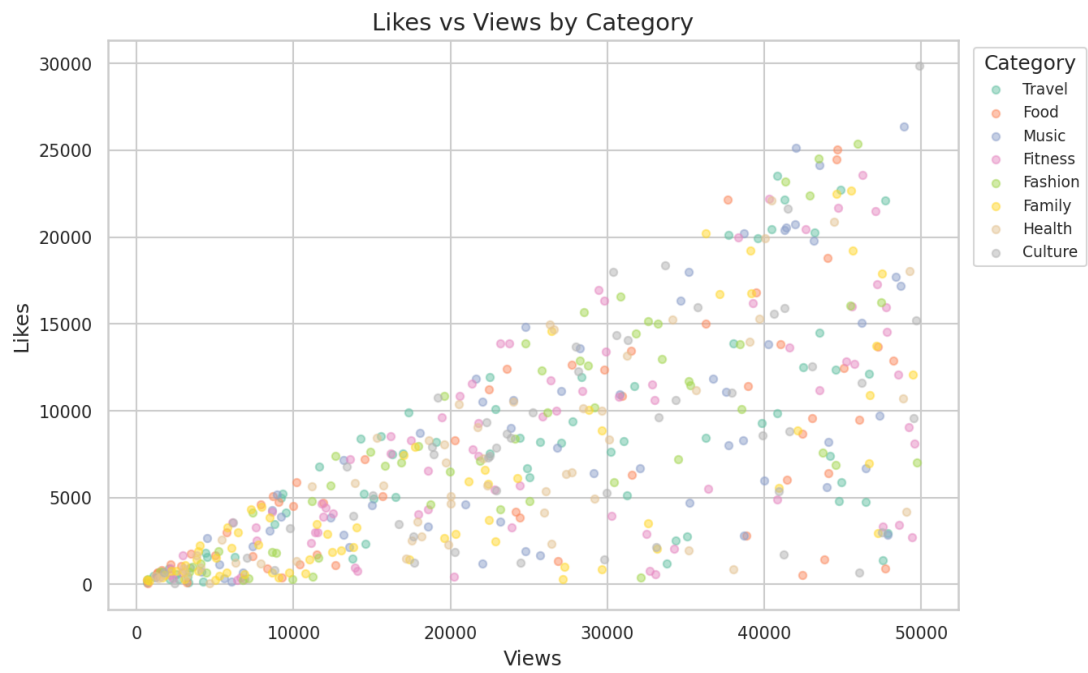


Figure 7 — Likes vs Views by Category

A positive linear trend exists between Views and Likes across all categories. The colour coding confirms that no single category dominates the high-views space, and category-level scatter is consistent, supporting the engagement rate findings.

5. Statistical Findings

5.1 Overall Engagement Rate

Statistic	Value
Mean	40.17%
Std Dev	16.80%
Min	4.04%
25th percentile	27.40%
Median (50th)	41.18%
75th percentile	54.08%
Max	74.57%

5.2 Engagement Rate by Category

Category	Avg Engagement Rate (%)
Fashion	43.33
Food	41.18
Travel	41.14
Music	40.88
Fitness	40.70
Culture	39.53
Health	39.51
Family	35.59

5.3 Engagement Rate by Platform

Platform	Avg Engagement Rate (%)
Twitter	42.13
Facebook	40.63
Instagram	40.26
TikTok	38.12

5.4 Top 10 Posts by Total Engagement

Date	Category	Platform	Views	Engagement	Rate %
2023-12-03	Culture	Instagram	49,877	37,194	74.57
2023-03-11	Fitness	Facebook	46,258	30,596	66.14
2023-06-18	Fashion	TikTok	43,441	30,455	70.11
2024-05-07	Food	Facebook	44,578	29,097	65.27
2023-05-24	Music	Instagram	42,023	28,739	68.39
2023-12-23	Travel	Twitter	47,702	28,633	60.02
2024-01-23	Family	TikTok	45,517	28,453	62.51
2023-01-11	Food	TikTok	44,631	28,347	63.51
2023-12-04	Music	TikTok	48,904	28,248	57.76
2024-01-30	Health	Twitter	40,474	27,942	69.04

6. Key Insights

1. Fashion drives the highest engagement.

Fashion content averages a 43.33% engagement rate — 7.74 percentage points above the lowest category (Family at 35.59%). For a content team, this signals that Fashion-style posts warrant more investment.

2. More views do NOT mean better engagement rates.

The correlation between Views and engagement_rate is effectively zero ($r = -0.01$). Viral reach and proportional audience interaction are independent. A post with 5,000 views can outperform one with 50,000 views in engagement quality.

3. Likes are the strongest predictor of total engagement.

Likes and total_engagement correlate at $r = 0.98$. For quick reporting, Likes alone can serve as a reliable proxy for overall engagement without needing to sum Comments and Shares.

4. Platform differences are narrow but real.

Twitter leads at 42.13% vs TikTok at 38.12% — a 4pp gap. While statistically present, this spread suggests platform choice is secondary to content quality. A real-world analysis with more data could test this with a statistical significance test.

5. Top-performing posts are spread across all categories.

The Top 10 posts span 8 different category-platform combinations, confirming that no single pairing monopolises high engagement. This supports a diversified content strategy rather than over-indexing on one category.

7. Code Overview

The full script is structured as a 7-step pipeline. Each step is encapsulated in a named function with a docstring, following clean code principles. All charts are saved programmatically — no `plt.show()` calls are used, making the script suitable for server-side or CI/CD execution.

Dataset Generation — Realistic Ratios

```
views    = np.random.randint(500, 50_000, size=n)
likes    = (views * np.random.uniform(0.01, 0.60, size=n)).astype(int)
comments = (views * np.random.uniform(0.00, 0.10, size=n)).astype(int)
shares   = (views * np.random.uniform(0.00, 0.08, size=n)).astype(int)
```

Feature Engineering — Engagement Rate

```
df['total_engagement'] = df['Likes'] + df['Comments'] + df['Shares']
df['engagement_rate']  = (df['total_engagement'] / df['Views'] * 100).round(2)
```

Fix Applied — histplot replaces deprecated distplot

```
# DEPRECATED (crashes on Seaborn >= 0.12):
# sns.distplot(df['Likes'], bins=30, kde=True)

# CORRECT:
sns.histplot(df['Likes'], bins=30, kde=True, ax=ax, color='#4C72B0')
```

Statistical Analysis — Groupby

```
cat_stats = (df.groupby('Category')['engagement_rate']
              .mean()
              .sort_values(ascending=False)
              .round(2))
```

8. Conclusion

This project demonstrates a complete, professional-grade EDA pipeline on a social media dataset. From raw data generation through to visualisation and automated insight extraction, every stage follows real analyst best practices.

The most significant finding is the independence of reach and engagement quality: views volume and engagement rate are uncorrelated, meaning that content relevance drives proportional interaction — not algorithmic amplification alone. This is a nuanced, defensible business insight that goes beyond a simple 'Fashion gets more likes' observation.

Future Improvements

- Use a real-world dataset (e.g. Kaggle Instagram or Twitter engagement data)
- Convert to a Jupyter Notebook for interactive exploration and inline charts
- Build a Streamlit dashboard for non-technical stakeholders
- Add a simple ML regression model to predict engagement rate from post features
- Extend to multi-year analysis to identify seasonal engagement patterns
- Add statistical significance testing (t-test / ANOVA) for category comparisons